РЕКЛАМА

ИССЛЕДОВАТЕЛЬСКАЯ СТАТЬЯ

Марсианский метеорит, образовавшийся ≥4,1 миллиарда лет назад, содержит образцы ранней коры и мантии Марса

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Абстрактный

Марсианские метеориты хранят в себе геологическую историю планеты. Однако большинство из них — это геологически молодые образцы, которые не являются репрезентативными для поверхности Марса, где преобладают древние породы. Мы описываем марсианский метеорит Тегаза 001 (Teg 001) — медленно остывшую магматическую породу, образовавшуюся по меньшей мере 4,1 миллиарда лет назад. Его состав, минералогия и текстура отличаются от других марсианских метеоритов. Мы пришли к выводу, что Teg 001 — это образец древней коры, сформировавшейся из геохимически отличного мантийного источника с первичным составом. Наличие минералов, подвергшихся водному выветриванию, в сочетании с измерениями изотопов кислорода в образцах и минералах указывают на то, что Teg 001 вступал в реакцию с флюидами, обогащенными ^{18}O, до того, как был выброшен в космос в результате столкновения.

ПОДПИШИТЕСЬ НА УДОСТОЕННУЮ НАГРАД ИНФОРМАЦИОННУЮ РАССЫЛКУ *SCIENCE*ADVISER
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На Марсе не происходит тектоники плит в глобальном масштабе, поэтому на его поверхности сохранились геологические свидетельства более чем четырех миллиардов лет планетарной эволюции. Размер и плотность ударных кратеров указывают на то, что большая часть поверхности Марса древняя и сформировалась $\gtrsim$3,0 миллиарда лет назад (млрд лет) (*1*). Поверхность состоит из смеси осадочных и магматических пород.

Состав и эволюция коры и мантии Марса были установлены на основе геохимического анализа марсианских метеоритов — образцов планеты, которые естественным образом попали на Землю в результате столкновений. Однако большинство марсианских метеоритов представляют собой магматические породы и намного моложе большей части поверхности планеты. Три основных класса марсианских метеоритов — шерготтиты, нахлиты и шассиньиты (SNC) — образовались не более 2,4 млрд лет назад (*2*). Было обнаружено всего два древних марсианских метеорита: Allan Hills 84001 (ALH 84001) — мономинеральная порода, сформировавшаяся 4,1 млрд лет назад (*3*); и Northwest Africa

7034 (NWA 7034) и его парные брекчии (смеси различных минералов и фрагментов горных пород), в которых есть фрагменты, сформировавшиеся более 4,3 млрд лет назад (*4*–*6*). Эти два образца использовались для изучения ранней эволюции коры и мантии Марса (*3*, *7*, *8*), но они дают неполную информацию о магматических процессах и условиях на поверхности древнего Марса. Небольшое количество магматических метеоритов возрастом более 3 млрд лет свидетельствует о недостаточном изучении композиционного разнообразия ранней марсианской коры и мантии (*9*), что оставляет пробелы в наших представлениях о ранней геологической эволюции Марса.

Мы описываем Тегаза 001 (далее — Тег 001), древний магматический марсианский метеорит. Он был найден в пустыне Сахара в Мали в 2022 году (*10*), и его марсианское происхождение было подтверждено с помощью измерения содержания изотопов кислорода (таблица S1 и рис. S1). Тег 001 состоит из восьми камней общей массой 798,5 г (*11*), самый крупный из которых весит 442 г (рис. 1). Мы использовали несколько методов (*11*), чтобы определить его текстуру, минеральный состав, структуру и возраст.

Петрография, минералогия и состав

Teg 001 представляет собой среднезернистую магматическую породу, состоящую в основном из минералов силиката кальция: 68% по площади поверхности составляет плагиоклазовый полевой шпат [(Na, Ca)$Al_{1-2}Si_{2-3}O_8$] и 25,5% - клинопироксен [(Ca, Fe, Mg) $_2Si_2O_6$], последний включает пироксеновый пигеонит с низким содержанием Ca (20%) и пироксеновый авгит с высоким содержанием Ca (5,5%) (рис. 1C и инжир. S2 — S6). К второстепенным компонентам относятся мезостаз (остаточное стекло с высоким содержанием калия и кремния, рис. S4), фосфаты (рис. 1C и рис. S2 — S6), кремнеземная основная масса (рис. 2, A и B и рис. S5 и S6), оксиды железа и титана (в основном ильменит, $FeTiO_3$, рис. 2D и рис. S4A и S7), циркон ($ZrSiO_4$; рис. 2, C и D, и рис. S9) и другие акцессорные минералы (дополнительный текст).

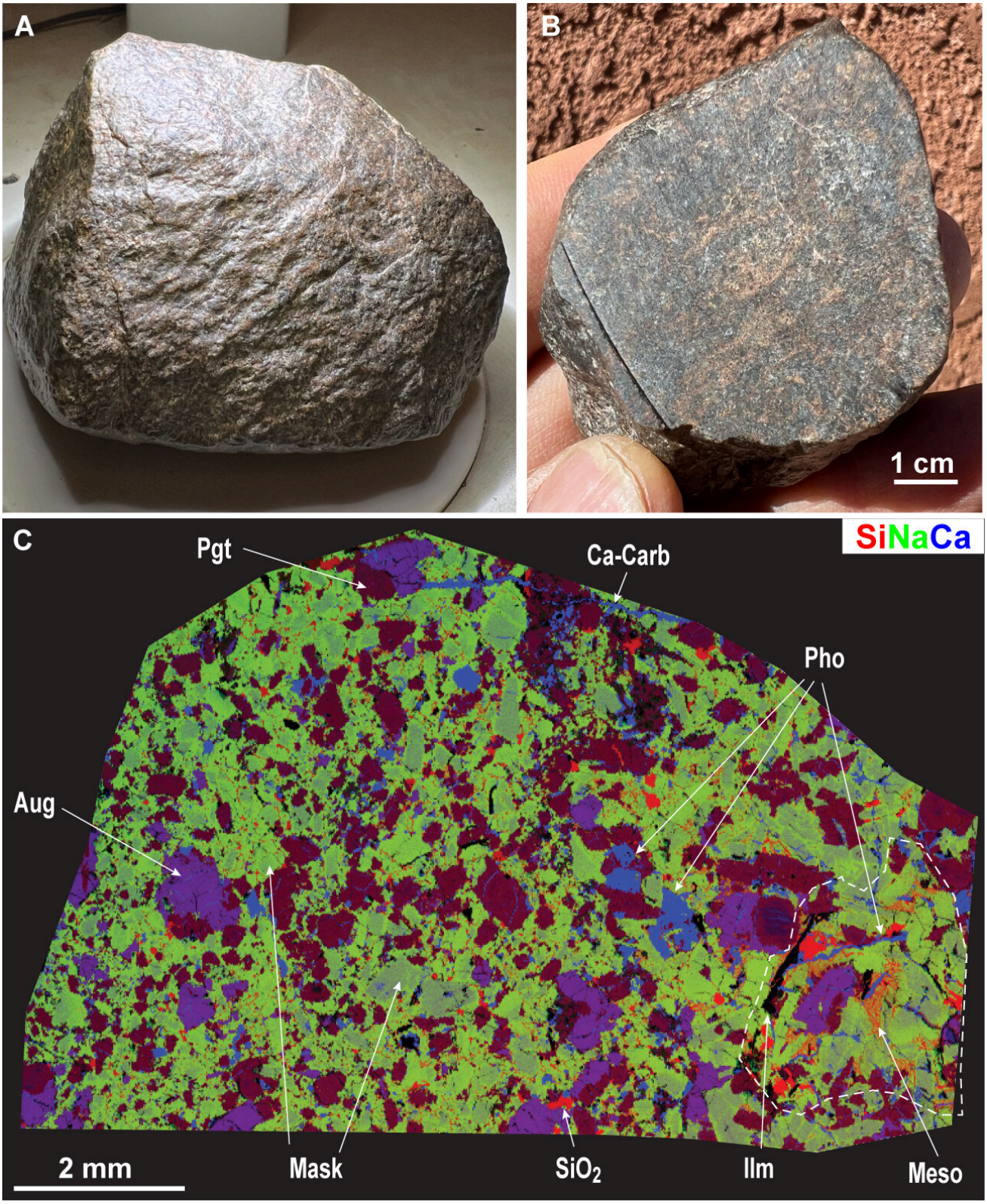


Рис. 1. Марсианский метеорит Тегаза 001.
(**A**) Фотография самого крупного образца (основной массы) Тега 001, который весит 442 грамма и имеет поперечное сечение ~10 см. (**B**) Фотография поверхности среза небольшого фрагмента. (**C**) Карта распределения элементов в полированном срезе Тег 001а. Красный цвет — кремний, зеленый — натрий, синий — кальций; на рис. S3 показаны отдельные изображения этих и других элементов в оттенках серого. Белыми стрелками обозначены конкретные минералы: Aug — авгит; Ca-Carb — карбонат кальция; Ilm — иль-

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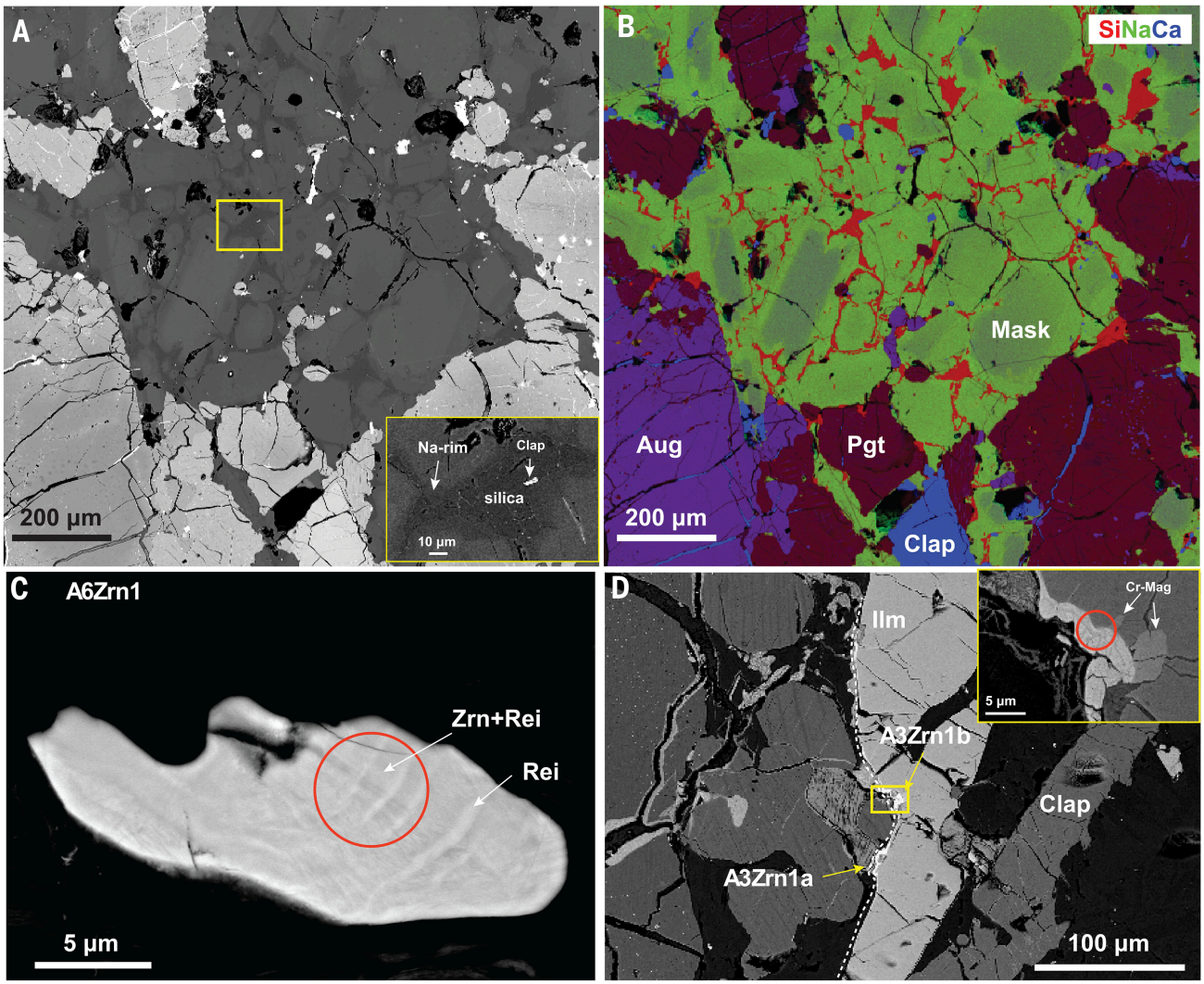


Fig. 2. Petrographic features of Teg 001.
(**A**) Backscattered electron (BSE) image showing the texture of major mineral phases. The yellow box marks the interstitial silica matrix shown in the inset. White arrows label the Na-rich rim (Na-rim) on the maskelynite and a small chlorapatite (Clap) grain in the matrix. (**B**) False-color element map of the same regions as in panel A, showing pervasive silica matrix. Colors are the same as Fig. 1C and fig. S5 shows grayscale images for each element. (**C**) BSE image of a partially formed zircon (designated A6Zrn1 in data S1). The surrounding area of this grain was

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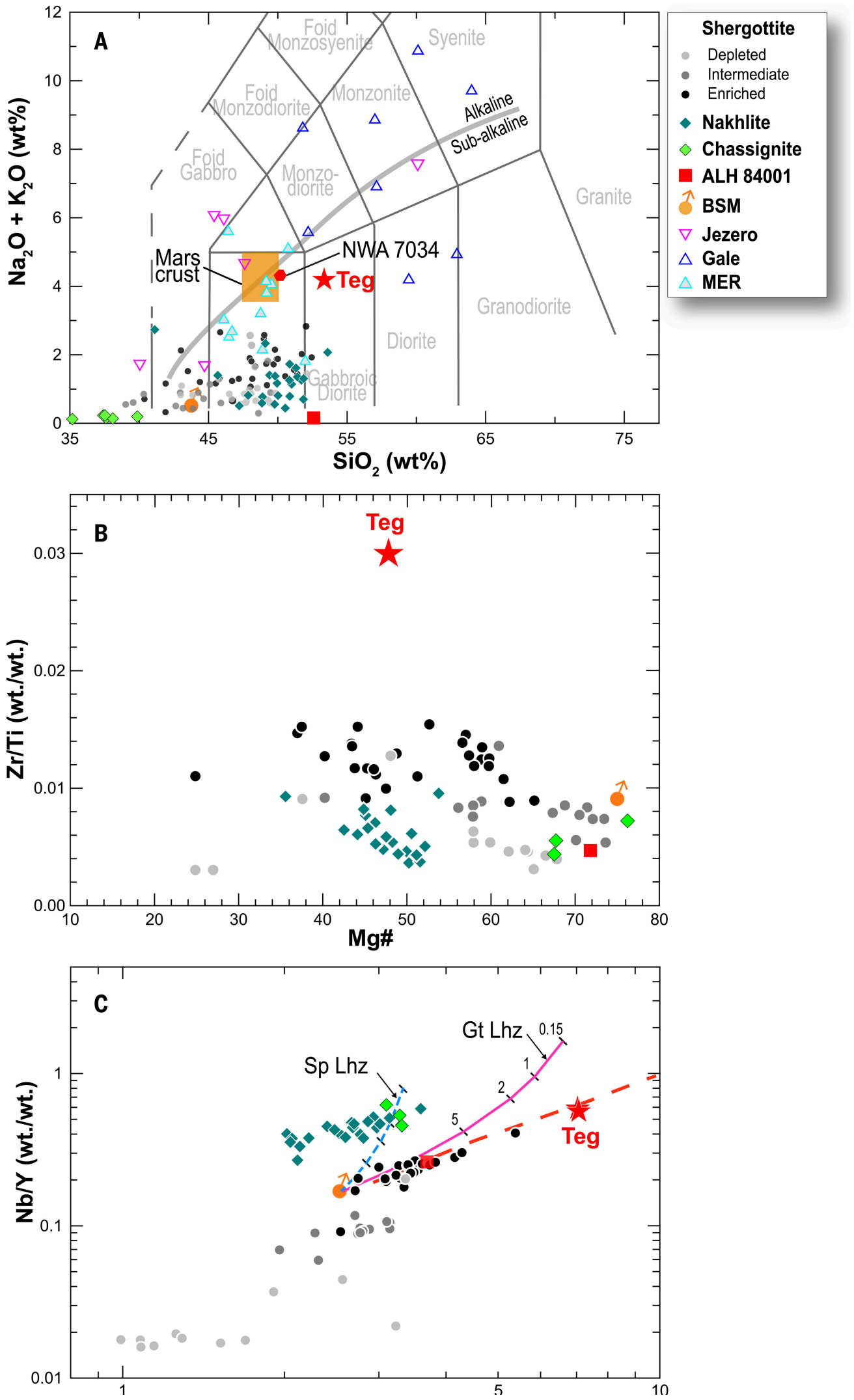
Alteration of Teg 001 on Earth has deposited calcium carbonate (Fig. 1C) and barium sulfate in fractures, and corroded Fe-sulfide to ferric iron oxide. This affected Ba and Sr abundances of Teg 001 but left other trace elements unaffected (supplementary text). Mineral abundances measured from a polished section (*11*) are consistent with values calculated from the bulk chemistry of a different piece, indicating representative sampling (tables S4 and S5). All the pyroxenes contain thin lamellae (figs. S7 and S8), which formed by solid-state exsolution while the rock cooled slowly at ~0.01°C year^{-1} (*12*), consistent with an intrusive origin. Using classification schemes for intrusive rocks (*13*, *14*), the bulk composition and mineralogy of Teg 001 indicate it is a leuco-gabbro (leuco indicates more plagioclase than pyroxene in the sample).

Shock metamorphism has transformed crystalline plagioclase to non-crystalline maskelynite (fig. S10), and some individual zircon grain into its high-pressure polymorph reidite (figs. S11 to S13). Formation of reidite requires a peak shock pressure of 10 to 20 GPa, and maskelynite formation is consistent with the same range (*15*). Post-shock heating did not reach >1200°C, otherwise reidite would have back-transformed to zircon or the impact would have produced shock-melting in the sample, which we do not observe (supplementary text).

Teg 001 displays textural, mineralogical, and compositional features that are unusual for Martian meteorites (supplementary text). Plagioclase grains contain Na-rich rims, 5 to 10 μm thick, which form a sharp contact with their Ca-rich interiors (Fig. 2, A and B, and figs. S5 and S6) and Na-rich zones associated with fractures (fig. S7A). These features indicate that the Na-rich rims formed after the plagioclase crystallized. The Mg-rich interiors of pigeonite have been heavily altered by reactions with a fluid phase, whereas the Fe-rich rims remain mostly pristine (fig. S8A). The shergottite NWA 7042 contains similar alteration in the interior of olivine grains, which formed on Mars (*16*). Phosphates occur in three morphologies: large polycrystalline aggregates, vein shapes, and slender single crystals (Fig. 1) (*17*). Interstitial areas between mineral grains consist of amorphous silica (Fig. 2, A to D, and figs. S5, S6, and S8), which is atypical for leuco-gabbro.

Zircon occurs as two different morphologies: partially formed (subhedral) grains ($\lesssim$5 μm × 10 μm) in the rock matrix (Fig. 2C), and string shapes at ilmenite grain edges (<1 μm × $\lesssim$500 μm, Fig. 2D and fig. S12). String zircon is atypical for zircon crystallization from a silicate melt. Both shapes are shocked and contain Na-rich regions with 0.3 to 2 wt% Na_2O (data S1). This high sodium content is uncommon in zircons and is likely a nanoscale byproduct of alteration on Mars, which was subsequently incorporated into the zircon structure by impact heating, similar to the incorporation of nano-magnetite into the zircons in NWA 7034 (*18*).

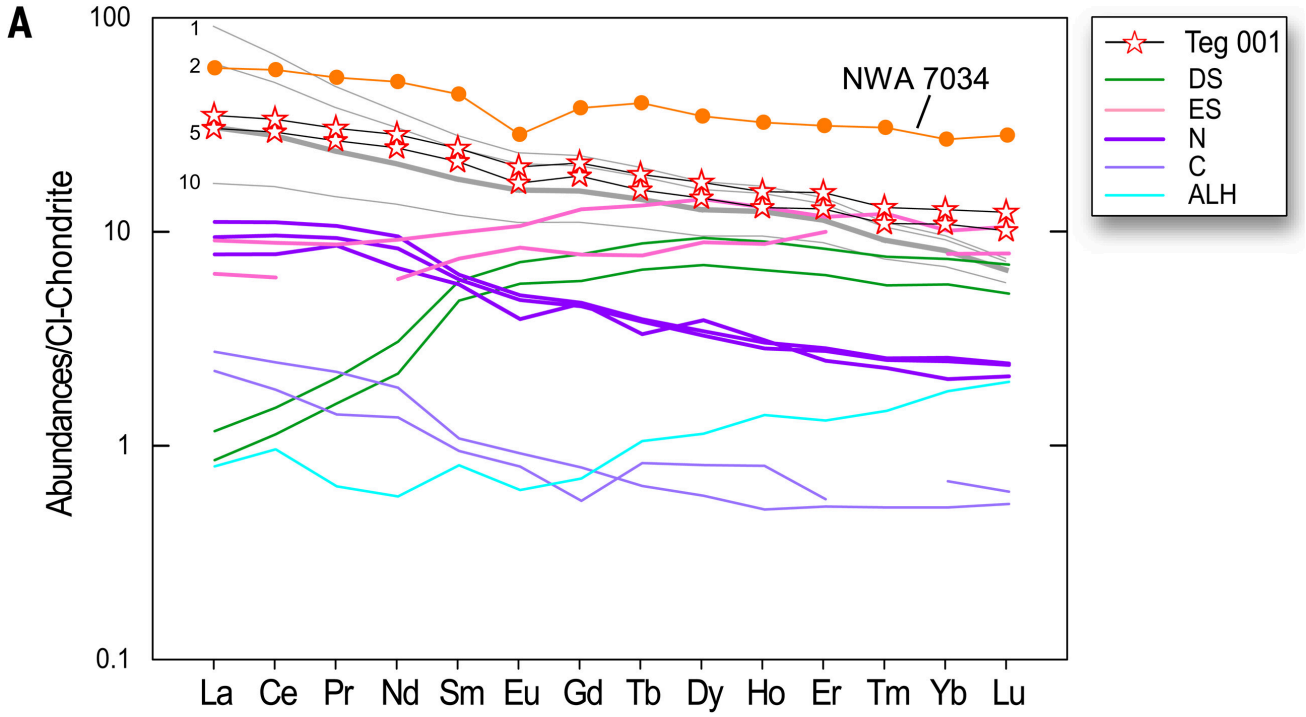
The bulk composition of Teg 001 differs from other igneous Martian samples (Fig. 3 and fig. S14). Trace elements that are incompatible in silicate minerals so concentrate in a silicate melt (incompatible elements) are ~10 to 60 times higher in Teg 001 than in Ivuna-type carbonaceous (CI) chondrite meteorites (taken to be the initial composition of the Solar System), except for Cs (~1.6 × the abundance in CIs), Rb (~7 × CI), and Pb (~0.3 × CI) (Fig. 4). The absolute abundances of trace elements in Teg 001 are higher than other igneous Martian meteorites, except for NWA 7034.


A
Foid Monzosyenite
Syenite
Foid Monzodiorite
Monzonite
Alkaline
Sub-alkaline
Foid Gabbro
Monzo-diorite
Granite
Mars crust
NWA 7034
Teg
Granodiorite
Diorite
Gabbroic Diorite
Na2O + K2O (wt%)
SiO2 (wt%)
Shergottite
Depleted
Intermediate
Enriched
Nakhlite
Chassignite
ALH 84001
BSM
Jezero
Gale
MER
B
Teg
Zr/Ti (wt./wt.)
Mg#
C
Gt Lhz
0.15
1
2
5
Sp Lhz
Teg
Nb/Y (wt./wt.)

Zr/Y (wt./wt.)

Fig. 3. Composition of Teg 001 compared to other Martian meteorites and orbital data.
(**A**) Rock classification using $Na_2O + K_2O$ content as a function of SiO_2. Data for other Mars samples and Teg 001 (red star) have been renormalized so the elemental oxides add to 100%. Thin gray lines are compositional boundaries of different rock types . The thick gray curve is the boundary between alkaline and subalkaline rocks . Other Martian meteorites: ALH 84001 (red square), chassignites (green diamonds), nakhlites (teal diamonds) and shergottites (black and gray circles) . For shergottites, we follow a previous classification as depleted (light

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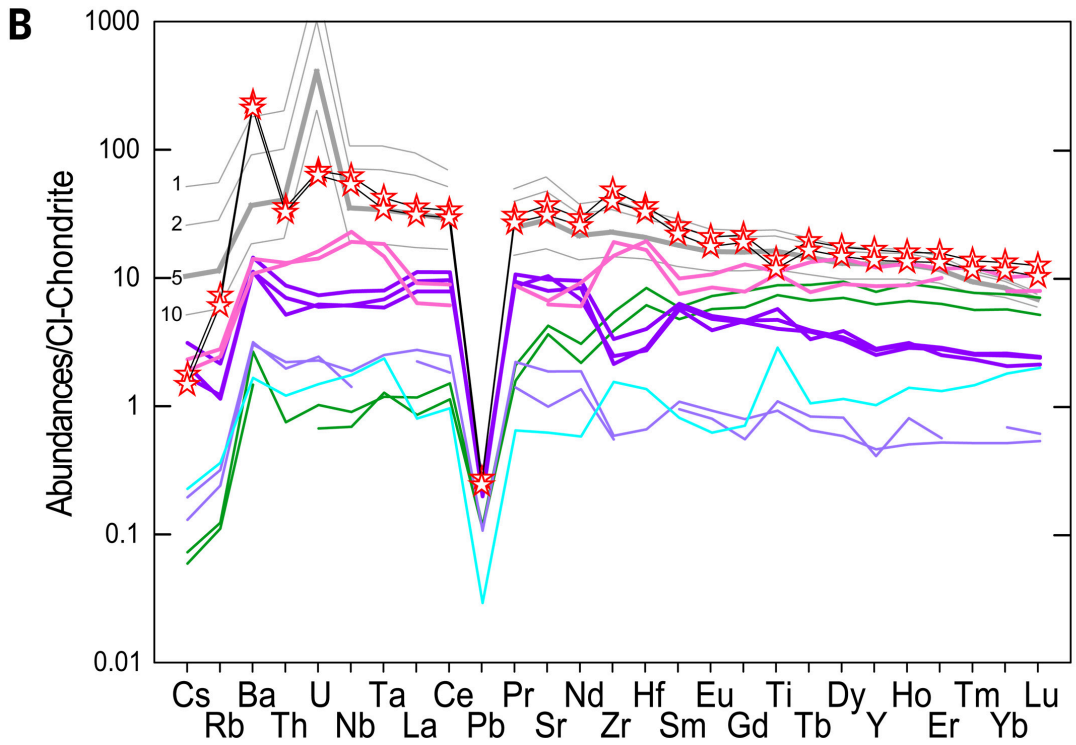
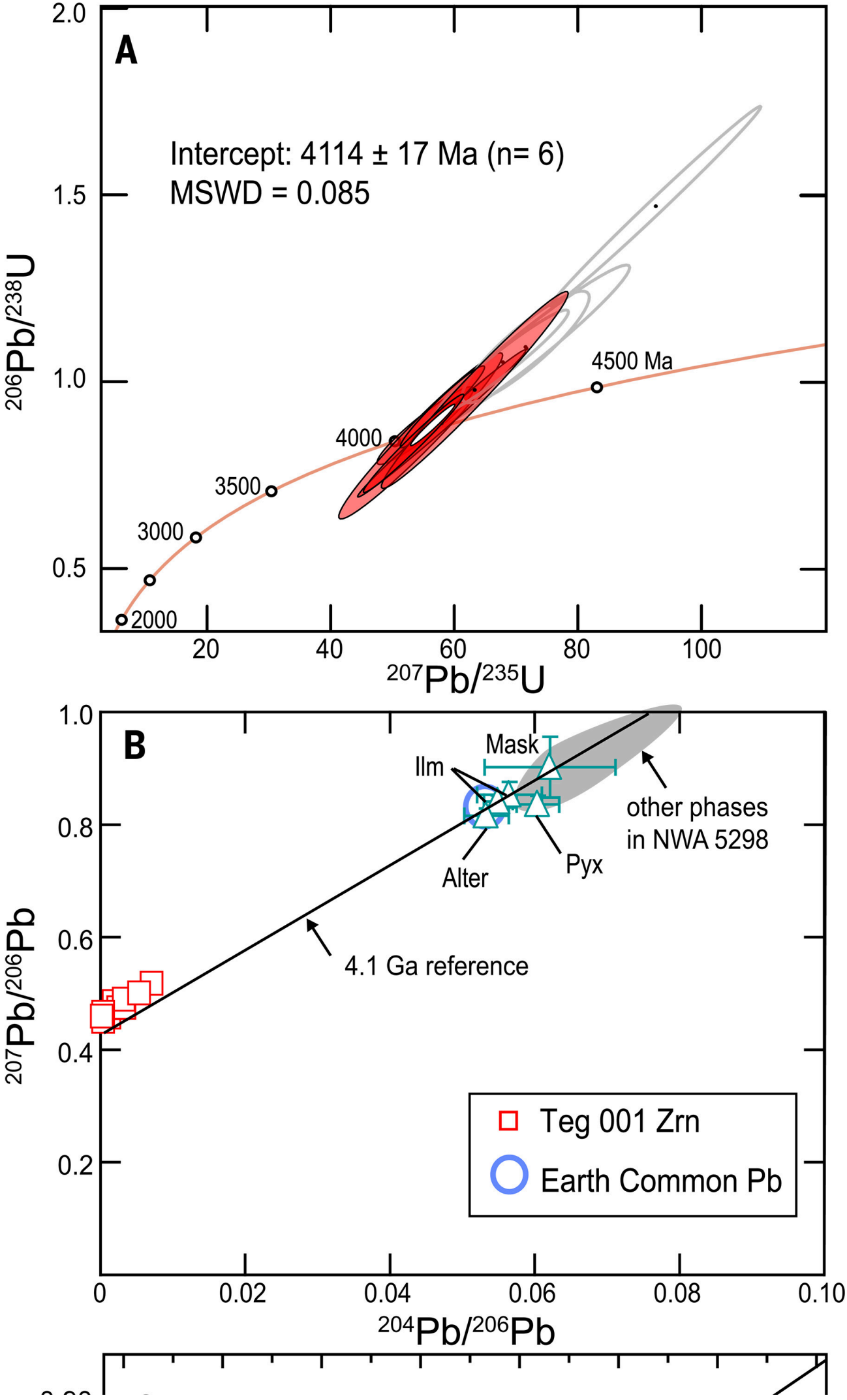


Fig. 4. Trace element abundances of Teg 001, Martian meteorite falls and ALH 84001.
Elemental abundances of (**A**) rare earth elements and (**B**) other trace elements, in decreasing order of their tendency to remain in the melt during crystallization. All measurements have been normalized to CI chondrites . Missing values were below the detection limit or not analyzed in previous studies. Red stars connected by thin black lines are two different analyses for Teg 001 . Gray lines are theoretical calculations for partial melting of garnet lherzolite, starting from bulk silicate Mars composition; each line shows a different degree of partial

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Dating and age interpretation

We conducted ten measurements of U and Pb isotopes (data S2), in six shocked zircon grains with different morphologies, to determine their U-Pb and Pb-Pb ages (*11*). Six measurements of $^{207}Pb/^{235}U$ and $^{206}Pb/^{238}U$ give similar ages (Fig. 5A), with a weighted average of 4114 ± 17 Ma (uncertainty given as two standard errors on the mean, 2 s.e). Four U-Pb measurements produce discrepant ages, due to sampling of neighboring phases (see supplementary text). Ten $^{207}Pb/^{206}Pb$ ages of zircon have a weighted average of 4113 ± 14 Ma (2 s.e.) (Fig. 5B). In situ Rb-Sr dating of plagioclase, pyroxene, and matrix (*11*) indicates an isochron age of 4238 ± 208 Ma (2 s.e.) and initial $^{87}Sr/^{86}Sr = 0.70270 \pm 0.00059$ (Fig. 5C). Previous measurements of eight $^{207}Pb/^{206}Pb$ ages of phosphates with different morphologies had a weighted average age of 4124 ± 28 Ma (*17*).


A
Intercept: 4114 ± 17 Ma (n= 6)
MSWD = 0.085
4500 Ma
4000
3500
3000
2000
$^{206}Pb/^{238}U$
$^{207}Pb/^{235}U$
B
Mask
Ilm
Alter
Pyx
other phases
in NWA 5298
4.1 Ga reference
Teg 001 Zrn
Earth Common Pb
$^{207}Pb/^{206}Pb$
$^{204}Pb/^{206}Pb$

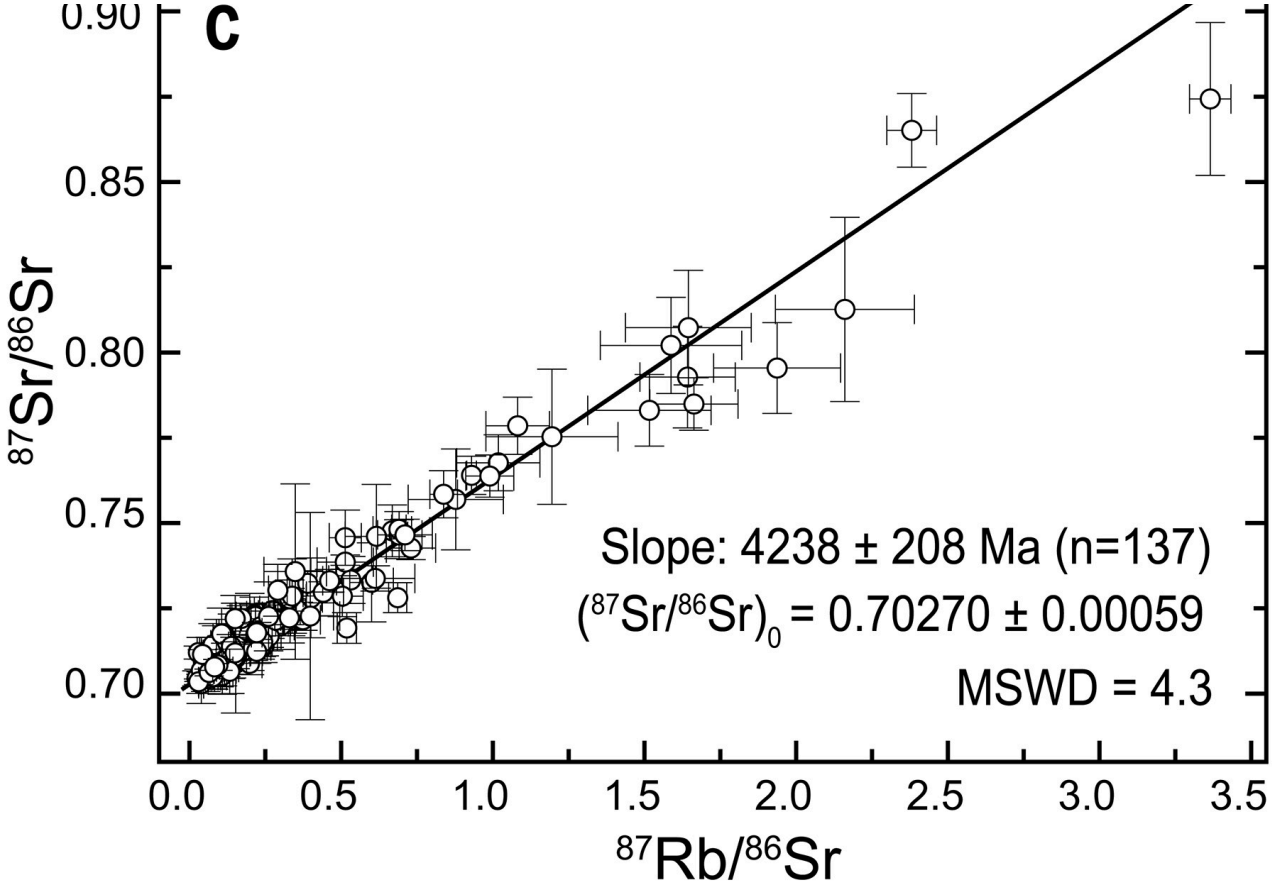


Fig. 5. Isotopic dating of Teg 001.
(**A**) Results for the U-Pb chronometer. The orange curve shows the expected $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ values (concordant curve) for samples with the labelled formation ages (in Ma). Gray ellipses show the 2σ uncertainty regions for our ten measurements of six zircon grains (data S2). Six measurements (pink shading) intersect the concordant curve, at a mean age of 4114 ± 17 Ma. Four measurements (empty ellipses) lie above the concordant curve (so are reverse discordant), indicating either imperfect matrix correction, the addition of non-radiogenic Pb, or the

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All these chronometers are consistent with an age of ~4.1 Ga. This date could reflect the time of magma crystallization or a subsequent event, such as an impact or an episode of alteration that reset the chronometers. The limited shock heating inferred above (from the presence of reidite and the lack of impact melt) is insufficient for complete Pb loss across multiple zircon grains of different sizes, due to the slow diffusion rate of Pb in zircon (*19*). Laboratory shock experiments and analyses of Earth impact ejecta have shown that shock deformation alone does not affect the U-Pb and Pb-Pb isotopic systematics in zircons (*20*). Consequently, the Na-rich alteration in zircon and plagioclase indicates that ~4.1 Ga is the date of hydrothermal alteration. The concordance between the $^{207}Pb/^{235}U$ and $^{206}Pb/^{238}U$ ages indicates that this hydrothermal event occurred shortly after primary crystallization (supplementary text). Teg 001 therefore formed no later than ~4.1 Ga, during the pre-Noachian geological period of Mars.

Formation and evolution

The mineralogy, texture, and composition of Teg 001 are consistent with it having initially formed by crystallization of a parent melt, derived from the Martian mantle (supplementary text). After crystallization, Teg 001 experienced protracted cooling, as indicated by the exsolution features in pyroxene. The string zircon (Fig. 2D and fig. S1, A to C) might also have formed during this cooling stage, by reaction of residual melt with Zr diffusing out of ilmenite, as in mafic layered intrusions on Earth (supplementary text).

Teg 001 experienced at least one subsequent Na-rich alteration event, while still on Mars. The ubiquitous Na enrichment in both zircon morphologies (data S1) indicates that alteration occurred after string-zircon formation.

The presence of Na-rich reidite (data S1) indicates that alteration occurred prior to shock transformation of zircon. Contamination by Earth weathering is unlikely, because zircon is resistant to surface weathering and Na-rich alteration of zircon is not a typical surface weathering product on Earth (*21*, *22*). The Na-rich rims on plagioclase might have been formed by the same alteration event that affected the zircons. The timing of the alteration of pigeonite interiors is unclear and could be from a different process.

Enrichment in the isotope ^{18}O is quantified by $\delta^{18}O$ (defined in equation S1). The average bulk value in Teg 001 is $\delta^{18}O = 6.35 \pm 0.15$ ‰ (uncertainty is one standard deviation, 1 s.d.) (table S1), which is ~2 ‰ higher than the average of SNC meteorites and ALH 84001 [4.46 ± 0.39 ‰ (*4*)]. Fractional crystallization of dry basaltic melt only increases the $\delta^{18}O$ of residual melt by <1 ‰ as the melt SiO_2 increases from ~45 to 55 wt% (*23*). Magmatic degassing of SO_2, H_2O or CO_2 lowers the residual melt $\delta^{18}O$ by $\lesssim$ 0.35‰ (*24*). Therefore, neither crystallization nor degassing could explain the 2 ‰ difference. The high $\delta^{18}O$ value of Teg 001 probably reflects assimilation of $\delta^{18}O$-rich crustal material that had interacted with surficial fluids (*7*), or direct reaction of the magma with high-$\delta^{18}O$ fluids. Previous H isotope measurements of phosphates in Teg 001 showed evidence for interactions between the magma and isotopically enriched crustal fluids (*17*). In-situ $\delta^{18}O$ measurements of amorphous silica and K-rich glass in mesostasis range from 7 ± 0.7 ‰ to 10 ± 0.7 ‰ (1 s.d.). Equivalent measurements of zircons range from 7.0 ± 2.7 ‰ to 16.3 ± 2.8 ‰ (1 s.d.) (fig. S18). These high and heterogeneous values are consistent with direct reaction with the fluids.

At an unknown later time, Teg 001 was ejected into space by an impact, inducing a peak shock pressure of 20 GPa and heating it to <1200°C. This produced the high-pressure phases reidite and maskelynite, discussed above.

A sample of the ancient crust

The K/Th ratio of Teg 001 is 4960 ± 329 by weight, which is close to the average of Martian crust [5300, no uncertainty given (*25*)]. We compare our bulk K, Th, Si, and Fe measurements of Teg 001 to remote sensing measurements to determine possible source terranes (*11*). The most likely location is in the Noachian southern highlands (*11*), particularly the region northwest of Hellas impact crater (fig. S19), where plagioclase-rich rocks have been observed (*26*). Orbital observations have shown that magmatic rocks in Noachian crust are dominated by plagioclase and low-Ca pyroxene (*9*, *27*). Therefore, we consider Teg 001 to be a sample of ancient crust.

Teg 001 extends the geochemical diversity of Martian igneous meteorites (*28*–*30*) and surface rocks investigated by landers and rovers (*31*–*33*). Figure 3A compares the composition of Teg 001 and other Mars samples to a rock classification scheme (*13*, *39*). Teg 001, young SNC meteorites, and two Noachian rocks from Gale crater, are all classified as sub-alkaline. The other Mars surface igneous rocks are classified as alkaline. This is consistent with previous inferences that Mars' early crust has two evolutionary trends (*9*). On Earth, alkaline rocks are affiliated with plume related magmatism, while sub-alkaline rocks are from subduction or water-rich melting. Compositional trends in other elements show that Teg 001 is intermediate between these two suites (figs. S14 and S16).

Constraints on the mantle source

The ratios of incompatible trace elements in igneous rocks are characteristic of specific mantle sources, because they are unaffected by crystallization or accumulation of early-formed minerals (such as olivine and orthopyroxene). Elements with small ionic radii and high valence states, such as Nb, Ta, Zr, Ti, and Hf, are more resistant to modification by alteration and metamorphism than fluid-mobile trace elements (such as Cs, Rb and Sr) (*34*). Teg 001 has higher Zr/Ti, Zr/Y, Zr/Hf, Nb/Ta, Ce/Pb, and U/Pb ratios than other Martian meteorites (Fig. 3, B and C, and fig. S16), indicating that the mantle source of Teg 001 was different from those of other Martian meteorites (supplementary text).

We conducted equilibrium melting calculations (*11*) to assess the mantle source mineralogy, based on the trace element variations. We assumed a composition of bulk silicate Mars (*35*) (fig. S15, C and D) and calculated trace elements in partial melts from two mantle rock types (Fig. 4). The calculations at 5% melting reproduce the observed bulk trace element pattern in Teg 001, with the closest match having garnet in the source (Fig. 4). Fractional melting of a mantle source containing garnet (*34*) also reproduces the Nb/Y and Zr/Y systematics (Fig. 3B). This implies that the mantle source of Teg 001 had primordial composition, so its incompatible elements were

not depleted by prior melt extraction. Garnet is a stable mineral in the Martian mantle at pressures ~2 to 9 GPa, so the melting occurred at $\gtrsim$160 km depth (*36*). Previous studies of ALH 84001 and SNC meteorites have proposed that Martian magma ocean crystallized into three to five mantle reservoirs at ~4.52 Ga, each of which displays variable degrees of depletion of incompatible elements (*3*, *37*). The primordial mantle source of Teg 001 was present at >4.1 Ga and is geochemically distinct from previously identified mantle reservoirs, indicating that early differentiation processes on Mars produced a highly heterogeneous mantle.

Implications for temporal igneous evolution

Orbital observations of <3.7 Ga surface igneous rocks on Mars have shown that SiO_2 decreases while Th increases over time (*38*). This trend was interpreted as arising from progressively deeper melting, due to secular cooling and thickening of the crust and non-convective mantle (*38*). This trend implies that igneous rocks from >3.7 Ga would contain even higher SiO_2 and lower Th, which is not supported by our measurements of Teg 001 (fig. S21).

Comparing with younger SNC meteorites, the ratio of low-Ca pyroxene to total pyroxene (~0.8 by volume) and the K content (0.4 wt%) of the >4.1 Ga Teg 001, are consistent with a trend of decreasing low-Ca pyroxene abundances and K contents of igneous rocks over time (*27*). Such trends might reflect increasing depth of melting, decreasing H_2O contents in mantle source regions, changing compositions of the source regions, or changes in the degree of partial melting over time. However, the formation of different mantle reservoirs at >4.1 Ga is inconsistent with a simple temporal variation of the mantle.

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Author contributions:

Conceptualization: Y.L., Investigation: Y.L., C.M., Y.G., E.A.B., H.L., A.M.C., H.M., T.E.Y., F.L.H.T. K.Z., J.H., L.M.S., C.B.A., Visualization: Y.L., L.S., A.M.C., Funding acquisition: Y.L., Project administration: Y.L., Supervision: Y.L., Writing – original draft: Y.L., Writing – review & editing: Y.L., L.S., J.H., C.M., E.A.B., Y.G., A.M.C., H.L., C.E.B., K.Z., C.B.A. F.L.H.T.

Competing interests:

The authors declare that they have no competing interests.

Data, code, and materials availability:

The 22 g classification sample of Teg 001 is held by the Institute of Meteoritics, University of New Mexico, USA. Requests to access this sample can be submitted to Beth Ha beth3ha@unm.edu and will be evaluated on a case-by-case basis for scientific merit of the proposed research. The 442 g main mass of Teg 001 and additional stones are held by the Maine Mineral & Gem Museum, Bethel, USA. Sample access requests can be submitted via https://mainemineralmuseum.org/discover/research-requests/; they will be evaluated (on a case-by-case basis) for the scientific merit of the proposed research. Our measured oxygen isotopes, bulk chemistry, mineral modal percentage and CIPW results, U-Pb isotopes, Neoma MS/MS setup and Rb-Sr isotopes are listed in tables S1 to S8. The mineral and glass composition measurements are available in data S1. The U-Pb and Pb-Pb dating results are available in data S2. The results from our equilibrium melting calculation are listed in data S3.

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Supplementary Materials

The PDF file includes:

Materials and Methods

Supplementary Text

Figs. S1 to S26

Tables S1 to S11

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Other Supplementary Material for this manuscript includes the following:

Data S1 to S3

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